

Homework 1: Solutions

STT 861 — Theory of Probability and Statistics I

Casella and Berger, *Statistical Inference*, Chapter 1, pp. 37–40

Homework problems: 1–11. Total: 39 points.

Problem 1

5 points

Verify the four set identities below.

Solution

(a) $A \setminus B = A \setminus (A \cap B) = A \cap B^c$. For any $x \in S$,

$$x \in A \setminus B \iff x \in A \text{ and } x \notin B \iff x \in A \cap B^c.$$

Moreover, once $x \in A$, the condition $x \notin A \cap B$ is equivalent to $x \notin B$. Thus $A \setminus (A \cap B) = A \setminus B$ as well.

(b) $B = (B \cap A) \cup (B \cap A^c)$. The distributive law gives

$$(B \cap A) \cup (B \cap A^c) = B \cap (A \cup A^c) = B \cap S = B.$$

The two pieces are disjoint, since $A \cap A^c = \emptyset$.

(c) $B \setminus A = B \cap A^c$. For any $x \in S$,

$$x \in B \setminus A \iff x \in B \text{ and } x \notin A \iff x \in B \cap A^c.$$

(d) $A \cup B = A \cup (B \cap A^c)$. Using part (b) and $B \cap A \subseteq A$,

$$A \cup B = A \cup (B \cap A) \cup (B \cap A^c) = A \cup (B \cap A^c).$$

Problem 2

5 points

Prove commutativity, associativity, and De Morgan's laws.

Solution

Fix an arbitrary $x \in S$. Equality follows in each case from equivalent membership conditions.

(a) **Commutativity.**

$$\begin{aligned} x \in A \cup B &\iff (x \in A \text{ or } x \in B) \iff x \in B \cup A, \\ x \in A \cap B &\iff (x \in A \text{ and } x \in B) \iff x \in B \cap A. \end{aligned}$$

(b) **Associativity.** An element belongs to $A \cup (B \cup C)$ exactly when it belongs to at least one of A, B, C . This is also the membership condition for $(A \cup B) \cup C$. Thus

$$A \cup (B \cup C) = (A \cup B) \cup C.$$

Likewise, membership in either $A \cap (B \cap C)$ or $(A \cap B) \cap C$ means membership in all three sets, so

$$A \cap (B \cap C) = (A \cap B) \cap C.$$

(c) **De Morgan's laws.**

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \text{ and } x \notin B \iff x \in A^c \cap B^c, \\ x \in (A \cap B)^c &\iff x \notin A \text{ or } x \notin B \iff x \in A^c \cup B^c. \end{aligned}$$

Therefore $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Problem 3**5 points**

State and prove De Morgan's laws for a finite collection of sets.

Solution

For $A_1, \dots, A_n \subseteq S$ and any integer $n \geq 1$,

$$\boxed{\left(\bigcup_{i=1}^n A_i\right)^c = \bigcap_{i=1}^n A_i^c}, \quad \boxed{\left(\bigcap_{i=1}^n A_i\right)^c = \bigcup_{i=1}^n A_i^c}.$$

Proof by induction. For $n = 1$, both identities are immediate. Suppose both identities hold for n . Applying the two-set laws from Problem 2 and then the induction hypothesis gives

$$\begin{aligned} \left(\bigcup_{i=1}^{n+1} A_i\right)^c &= \left[\left(\bigcup_{i=1}^n A_i\right) \cup A_{n+1}\right]^c \\ &= \left(\bigcup_{i=1}^n A_i\right)^c \cap A_{n+1}^c = \bigcap_{i=1}^{n+1} A_i^c, \end{aligned}$$

and

$$\begin{aligned} \left(\bigcap_{i=1}^{n+1} A_i\right)^c &= \left[\left(\bigcap_{i=1}^n A_i\right) \cap A_{n+1}\right]^c \\ &= \left(\bigcap_{i=1}^n A_i\right)^c \cup A_{n+1}^c = \bigcup_{i=1}^{n+1} A_i^c. \end{aligned}$$

This completes the induction for both identities. □

Problem 4**3 points**

Verify three constructions of sigma algebras.

Solution

Recall that a collection $\mathcal{B}$ of subsets of S is a sigma algebra if it contains S , is closed under complements relative to S , and is closed under countable unions.

(a) $\mathcal{B} = \{\emptyset, S\}$. This collection contains S . Complements interchange $\emptyset$ and S . A countable union of its members is $\emptyset$ if every member is empty, and is S otherwise. Thus all three requirements hold.

(b) $\mathcal{B} = \mathcal{P}(S)$, the collection of all subsets of S . Clearly $S \in \mathcal{P}(S)$. The complement in S of a subset of S is again a subset of S , and any countable union of subsets of S is a subset of S . Hence $\mathcal{P}(S)$ is a sigma algebra.

(c) Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be sigma algebras on the same space S , and put $\mathcal{B} = \mathcal{B}_1 \cap \mathcal{B}_2$. Both collections contain S , so $S \in \mathcal{B}$. If $A \in \mathcal{B}$, then A belongs to both sigma algebras; therefore A^c belongs to both and hence to $\mathcal{B}$. Finally, if $A_k \in \mathcal{B}$ for every $k \geq 1$, closure in each sigma algebra implies

$$\bigcup_{k=1}^{\infty} A_k \in \mathcal{B}_1 \quad \text{and} \quad \bigcup_{k=1}^{\infty} A_k \in \mathcal{B}_2.$$

The union is consequently in $\mathcal{B}$, proving that the intersection is a sigma algebra.

Problem 5**3 points**

Count possible initials with one surname and the specified numbers of given names. Use 26 letters, allow repeated letters, and retain the order of the initials.

Solution

(a) **Exactly two given names.** There are three initial positions, each with 26 choices. The multiplication principle gives

$$26^3.$$

(b) **Either one or two given names.** The two possible lengths are disjoint cases, giving

$$26^2 + 26^3.$$

(c) **Either one, two, or three given names.** Similarly,

$$26^2 + 26^3 + 26^4.$$

Problem 6**3 points**

Count domino pieces labeled by unordered pairs from $\{1, \dots, n\}$, allowing equal labels.

Solution

There are $\binom{n}{2}$ pieces with two distinct labels and n doubles. These cases are disjoint, so the number of pieces is

$$\binom{n}{2} + n = \frac{n(n+1)}{2}.$$

Problem 7**3 points**

Place n balls at random into n cells and find the probability that exactly one cell is empty.

Solution

Use the standard model: the balls are distinguishable, and each independently chooses one of the n labeled cells with probability $1/n$. Thus there are n^n equally likely assignments.

For $n \geq 2$, exactly one empty cell leaves $n - 1$ occupied cells. Since there are n balls, exactly one occupied cell must contain two balls, and all other occupied cells must contain one. Count the favorable assignments:

1. Choose the empty cell: n choices.
2. Choose the cell with two balls: $n - 1$ choices.
3. Choose those two balls: $\binom{n}{2}$ choices.
4. Assign the remaining balls bijectively to the remaining cells: $(n - 2)!$ choices.

Each favorable assignment determines these choices uniquely. Therefore

$$\mathbb{P}(\text{exactly one empty cell}) = \frac{n(n-1)\binom{n}{2}(n-2)!}{n^n} = \frac{\binom{n}{2}n!}{n^n}.$$

Problem 8**3 points**

Distribute 12 telephone calls among seven days and find the probability of at least one call on every day.

Solution

Assume that the 12 calls are independent and that each call is equally likely to occur on any of the seven days. For $i = 1, \dots, 7$, define

$$E_i = \{\text{day } i \text{ receives no calls}\}.$$

Then E_i^c means that day i receives at least one call, so the event we want is $\bigcap_{i=1}^7 E_i^c$.

Step 1: Express the desired event as a complement. By De Morgan's law,

$$\bigcap_{i=1}^7 E_i^c = \left(\bigcup_{i=1}^7 E_i \right)^c, \quad \mathbb{P} \left(\bigcap_{i=1}^7 E_i^c \right) = 1 - \mathbb{P} \left(\bigcup_{i=1}^7 E_i \right).$$

Here $\bigcup_{i=1}^7 E_i$ is the event that at least one day receives no calls.

Step 2: Expand the union using inclusion–exclusion. Subtracting the inclusion–exclusion formula for the union from 1 gives

$$\begin{aligned} \mathbb{P} \left(\bigcap_{i=1}^7 E_i^c \right) &= 1 - \sum_{i=1}^7 \mathbb{P}(E_i) \\ &\quad + \sum_{1 \leq i_1 < i_2 \leq 7} \mathbb{P}(E_{i_1} \cap E_{i_2}) \\ &\quad - \sum_{1 \leq i_1 < i_2 < i_3 \leq 7} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap E_{i_3}) \\ &\quad + \sum_{1 \leq i_1 < i_2 < i_3 < i_4 \leq 7} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap E_{i_3} \cap E_{i_4}) \\ &\quad - \sum_{1 \leq i_1 < \dots < i_5 \leq 7} \mathbb{P}(E_{i_1} \cap \dots \cap E_{i_5}) \\ &\quad + \sum_{1 \leq i_1 < \dots < i_6 \leq 7} \mathbb{P}(E_{i_1} \cap \dots \cap E_{i_6}) \\ &\quad - \mathbb{P}(E_1 \cap \dots \cap E_7). \end{aligned}$$

Step 3: Evaluate the intersection probabilities. Fix j distinct days $i_1, \dots, i_j$. The event $E_{i_1} \cap \dots \cap E_{i_j}$ means that all these j days are empty. Thus each of the 12 calls must fall on one of the remaining $7 - j$ days. For one call this has probability $(7 - j)/7$; independence of the calls gives

$$\mathbb{P}(E_{i_1} \cap \dots \cap E_{i_j}) = \left(\frac{7 - j}{7} \right)^{12}, \quad 1 \leq j \leq 7.$$

The remaining days are allowed to be empty as well: this intersection specifies that the chosen j days are empty, not that exactly j days are empty. There are $\binom{7}{j}$ ways to choose these days, and every such choice has the same probability. Consequently,

$$\sum_{1 \leq i_1 < \dots < i_j \leq 7} \mathbb{P}(E_{i_1} \cap \dots \cap E_{i_j}) = \binom{7}{j} \left(\frac{7 - j}{7} \right)^{12}.$$

For example, the single-day terms total $7(6/7)^{12}$, and the two-day terms total $\binom{7}{2}(5/7)^{12}$. When $j = 7$, the probability is zero, since all seven days cannot be empty when there are 12 calls.

Step 4: Substitute and simplify. Substituting these values into the expansion above yields

$$\mathbb{P}\left(\bigcap_{i=1}^7 E_i^c\right) = \sum_{j=0}^7 (-1)^j \binom{7}{j} \left(\frac{7-j}{7}\right)^{12} \approx \boxed{0.2285}.$$

Problem 9

3 points

Choose $2r$ shoes uniformly without replacement from n pairs, where $2r < n$. Find the probability of no matching pair.

Solution

There are $2n$ distinct shoes, so the sample space has $\binom{2n}{2r}$ equally likely subsets. To obtain no complete pair, the selected shoes must come from $2r$ different pairs. First choose these pairs in $\binom{n}{2r}$ ways; then choose one of the two shoes from each pair in 2^{2r} ways. Thus

$$\mathbb{P}(\text{no matching pair}) = \frac{\binom{n}{2r} 2^{2r}}{\binom{2n}{2r}}.$$

Each favorable subset is counted once because it uniquely specifies both the represented pairs and the selected shoe from each pair.

Problem 10

3 points

Draw calendar days uniformly without replacement from the 366 days of a leap year.

Solution

(a) The first 180 draws are evenly distributed among the months. Even distribution means exactly $180/12 = 15$ days from each month. There are seven 31-day months, four 30-day months, and one 29-day month. The unordered set of the first 180 draws is uniform over all $\binom{366}{180}$ subsets. Consequently,

$$\mathbb{P}(\text{15 days from each month}) = \frac{\binom{31}{15}^7 \binom{30}{15}^4 \binom{29}{15}}{\binom{366}{180}} \approx 1.6673 \times 10^{-9}.$$

The numerator chooses 15 dates separately from each month. No additional ordering factor is needed: each 180-date subset has the same number of possible draw orders.

(b) None of the first 30 draws is from September. September has 30 days, leaving $366 - 30 = 336$ allowable dates. Therefore

$$\mathbb{P}(\text{no September date in the first 30 draws}) = \frac{\binom{336}{30}}{\binom{366}{30}} \approx 0.0687.$$

Problem 11**3 points**

Two people each toss a fair coin n times. Find the probability that they obtain the same number of heads.

Solution

Assume all $2n$ tosses are independent. Let X and Y be the respective numbers of heads. Then X and Y are independent $\text{Binomial}(n, 1/2)$ random variables. The disjoint events $\{X = k, Y = k\}$, $k = 0, \dots, n$, give

$$\begin{aligned}\mathbb{P}(X = Y) &= \sum_{k=0}^n \mathbb{P}(X = k) \mathbb{P}(Y = k) \\ &= \frac{1}{4^n} \sum_{k=0}^n \binom{n}{k}^2.\end{aligned}$$

To evaluate the sum, use $\binom{n}{k} = \binom{n}{n-k}$ and the identity

$$\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}.$$

This identity counts ways to select n objects from two disjoint groups, each containing n objects: if k objects are selected from the first group, then $n - k$ must be selected from the second. Summing over all k counts every selection exactly once. Hence

$$\boxed{\mathbb{P}(X = Y) = \frac{\binom{2n}{n}}{4^n}}.$$